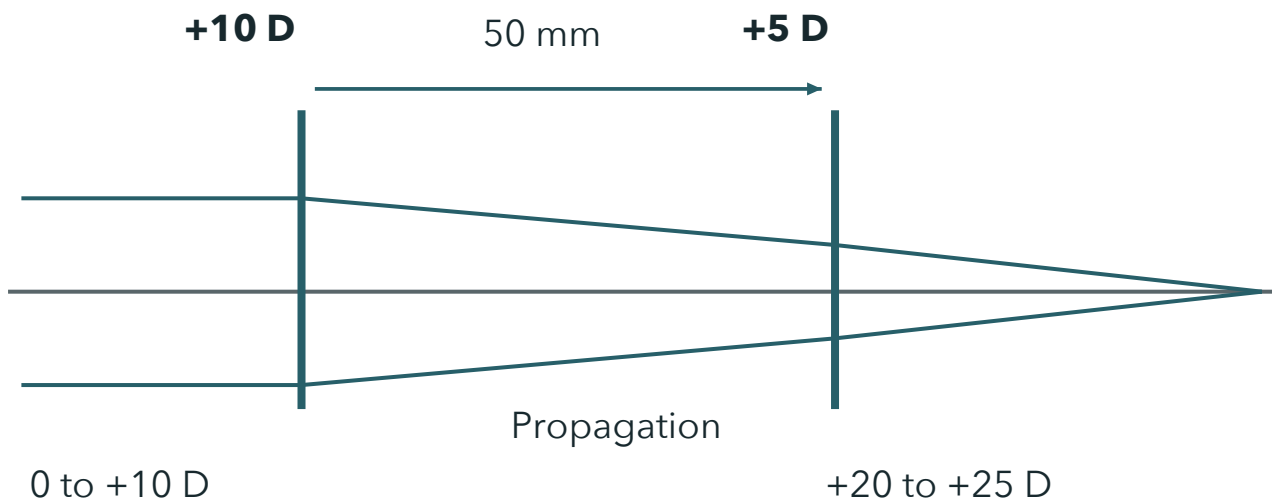


Vergence and cardinal planes

Part 1 | Propagate vergence between surfaces

Lens powers add at a surface; vergence changes again during travel to the next surface.



Two separated positive lenses are shown with propagation between the powers; the ray sketch is paraxial.

The two operations

Use $L_{\text{after}} = L_{\text{before}} + F$ at a thin lens and $L_{\text{next}} = L_{\text{after}} / [1 - dL_{\text{after}}]$ during propagation in air. Distance d is signed consistently and expressed in metres. A separated system is generally not equivalent to adding its powers without considering the separation.

A worked two-lens model

Track the vergence at each location instead of treating the system as one thin lens.

Collimated input; +10 D then +5 D, 50 mm apart

Location	Calculation	Vergence
First lens exit	$0+10$	+10 D
Second lens entrance	$10/(1-0.05 \times 10)$	+20 D
Second lens exit	$20+5$	+25 D

Check the sign

Converging light travelling toward its focus gains positive vergence before crossing that focus. If the denominator approaches zero, the next surface approaches the focal plane; beyond a focus the sign changes. State whether a focus is crossed before applying the intuitive direction.

Principal plane is not the lens surface

A focal length and a vertex distance can describe different reference points.

Locate the image

- Choose the posterior vertex as $x=0$
↓
- Place the second principal plane at its signed coordinate
↓
- Add the effective focal length measured from that plane
↓
- Report the image coordinate relative to the vertex

Original example

Let H_2 lie 3 mm anterior to the posterior vertex and the focal length from H_2 be 22 mm. The image of a distant object lies at $-3+22=19$ mm behind the vertex. A focal length of 22 mm is therefore not a 22 mm vertex focal distance.

A thick system has effective and vertex powers

The reference location matters even when the incident beam is the same.

Quantity

Feature	Reference	Useful distinction
Effective power	Principal-plane system	Describes system focusing
Back-vertex power	Posterior vertex	Important in spectacle measurement
Front-vertex power	Anterior vertex	Can differ from back-vertex value

Measurement interpretation

A focimeter reports a vertex-based quantity under its measurement conditions. Do not replace an effective focal length with a vertex focal length in a thick-system calculation unless the relevant planes coincide or their displacement has been accounted for.

Three safeguards for optical arithmetic

A correct formula with an inconsistent sign convention gives an incorrect image.

Check

Feature	Common failure	Repair
Units	Millimetres entered as metres	Convert before substitution
Location	Vergence taken at wrong surface	Draw each reference plane
Focus crossing	Sign carried through focus unchanged	Track the focal position

An examination method

Sketch the direction of travel and mark every optical element before calculating. Then annotate the vergence immediately before and after each element. This makes a missed propagation step visible and provides an independent plausibility check of the numerical answer.